

Scale specified single shot multibox detector

ISSN 1751-9632

Received on 15th June 2019

Revised 5th November 2019

Accepted on 20th November 2019

E-First on 23rd January 2020

doi: 10.1049/iet-cvi.2019.0461

www.ietdl.org

Xiaoqiang Li^{1,2} ✉, Chuanwei Liu¹, Songmin Dai¹, Huichen Lian¹, Guangtai Ding¹

¹School of Computer Engineering and Science, Shanghai University, Shanghai, People's Republic of China

²Shanghai Institute for Advanced Communication and Data Science, Shanghai University, Shanghai, People's Republic of China

✉ E-mail: xqli@shu.edu.cn

Abstract: Detecting objects at vastly different scales is a fundamental challenge in computer vision. To solve this, some approaches (e.g. TridentNet) investigate the effect of receptive fields, whereas other approaches (e.g. SNIP, SNIPER) are based on the image pyramid strategy. In this study, a novel single-shot based detector, called scale specified single-shot multibox detector (4SD) is proposed. It aims to predict objects of a specific scale range separately by using feature maps of different sizes. First, a parallel multi-branch architecture with feature maps of different sizes is generated by scale specific inference module. Then, the authors propose a scale specific training scheme to specialise each branch by sampling object instances of proper scales for training. Results are shown on both PASCAL VOC and COCO detection. The proposed method can achieve a mean average precision of 83.1% on PASCAL VOC 2007, and 36.9% on MS-COCO at a speed of 28 frames per second, which is superior to most single-stage detectors.

1 Introduction

In computer vision, detecting objects at vastly different scales is a fundamental challenge. The large-scale variation across object instances greatly impedes the detectors, especially for very small or very large objects. Image pyramids [1] form the basis of a standard solution for this challenge. In this way, an object's scale change is offset by moving its level in the image pyramid. They zoom the input image at multiple scales and then extract hand-crafted features, such as HOG [2], SIFT [3], from each image separately for detection. However, recent object detectors have avoided pyramid representations, in part because image pyramids are computation and memory intensive, like DPM [4] required dense scale sampling.

In recent years, deep learning [5, 6] has been successfully applied in the challenge of ImageNet classification [7], which has changed the way computers process pictures. The convolution neural network (CNN) can be used to extract the features representing higher-level semantics in object detection. Typically, these CNN-based methods can be roughly divided into two types: two-stage methods and single-stage methods. Two-stage methods, like Faster R-CNN [8], R-FCN [9], first generate region proposals and then exploit the extracted region features from CNNs for further refinement. Single-stage methods do not require a region proposal process, such as YOLO [10] which computes a global feature map and uses a fully-connected layer to predict the bounding boxes of interest, is significantly more efficient.

However, the feature map extracted by CNNs on a single input scale [8, 11–13] resulting in the fixed receptive field. To solve this, the single shot multibox detector (SSD) [14] uses feature pyramids to make prediction. Based on feature pyramids, TDM [15], DSSD [16] and FPN [17] further introduce a top-down pathway and lateral connections to enhance the semantic representation of low-level features at bottom layers. Imperfectly, CNNs are not robust to changes in scale. All aforementioned methods use the same shared inference module for all scales which decreases detection performance for very small or very large objects. SNIP [18] probes into the scale variance in object detection. To avoid scale mismatch between training and inference, SNIP proposes a scale normalisation method to train objects that fall into the desired scale range. TridentNet [19] generates scale-specific features through multiple parallel branches using dilated convolution [20] to endow

their network the same representational power for all objects of different scales.

Without applying the image pyramid strategy, the proposed scale specified SSD (4SD) takes a single-scale image as input. Different from TridentNet using dilated convolution to generate scale-specific feature maps, 4SD predict objects of a specific scale range separately by using feature maps of different sizes. 4SD contains shared feature extraction module connecting with a scale specific inference module (SSIM). The SSIM is constructed to generate multiple parallel branches with feature maps of different sizes to predict objects that fall into the desired scale range. To build SSIM, we upsample high-level feature map and enhance it with low-level feature map. Finally, we propose a scale specific training scheme (SSTS) to specialise each branch by sampling object instances of proper scales for training. Our model is a single-stage detector, and the detection speed is significantly faster than that of SNIP and TridentNet. Results are shown on both PASCAL VOC and COCO detection. 4SD can achieve a mean average precision (mAP) of 83.1% on PASCAL VOC 2007 [21], and 36.9% on MS-COCO [22] at a speed of 28 frames per second (FPS) on an NVIDIA 1080Ti GPU.

2 Related work

2.1 Deep CNN object detectors

With the development of deep CNNs, object detection methods, like OverFeat [23], R-CNN [24], have shown dramatic improvements in both accuracy and speed. OverFeat proposed by Sermanet *et al.* [23] adopt a strategy on an image pyramid. They apply CNNs as a sliding window detector and predict a bounding box directly from each location of the topmost feature map after knowing the confidences of the underlying object categories. R-CNN adopted a region proposal-based strategy [25]. Each proposal was scale-normalised before being classified using CNNs. SPPnet [11] demonstrated that such region proposal-based detectors could be applied much more efficiently on feature maps extracted on a single image scale. Faster R-CNN [8] introduces a region proposal network that shares the same backbone network with the detection network to replace the original standalone time-consuming region proposal methods. On the other hand, single-stage methods which are popularised by YOLO [10, 26, 27] and SSD [14], aim to be

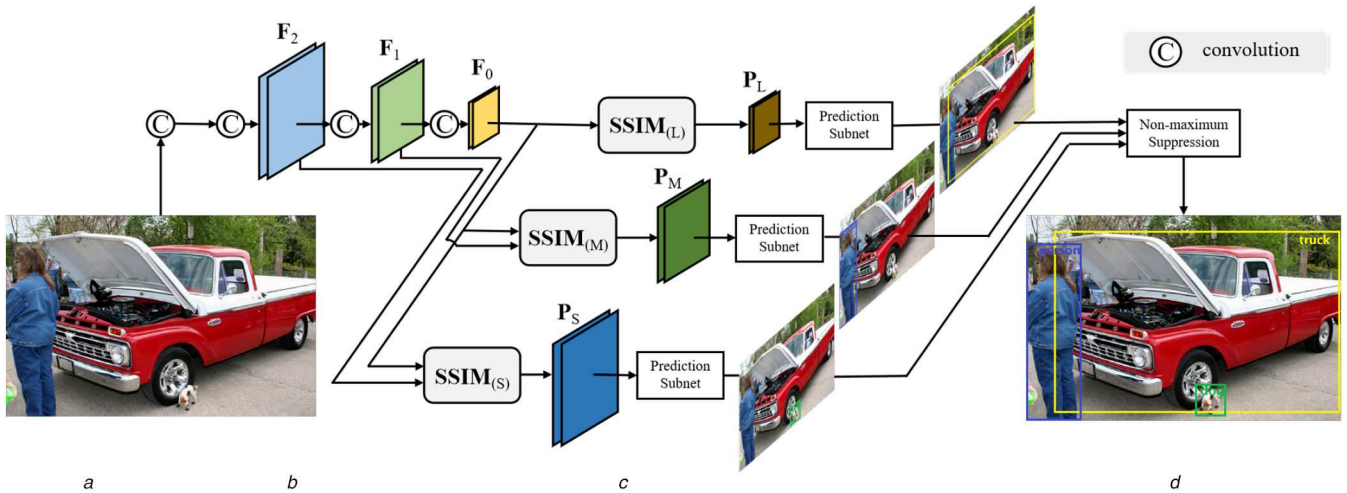


Fig. 1 Overview of scale specified single shot architecture

(a) Input image, (b) Base network is employed to obtain the input (F_2 , F_1 , and F_0) for SSIM, (c) SSIM is constructed to generate a parallel multi-branch architecture with feature maps of different sizes (P_S , P_M , and P_L). The desired scale object is predicted by the branch of feature maps of specific sizes, followed by, (d) Non-maximum suppression step to produce the final results

more efficient by directly classifying the pre-defined anchors and further refining them using CNNs without the proposal generation step. To alleviate the extreme foreground-background class imbalance encountered during training of dense single-stage detectors, RetinaNet [28] proposed a novel Focal Loss which focuses training on a sparse set of hard examples.

2.2 Methods using multiple layers

Image pyramids-based method enables a model to detect objects across a large range of scales by scanning the model over pyramid levels. However, the image pyramids-based method still suffers from the inevitable increase of inference time. Compared to take image pyramids strategy, methods using multi-level features in CNNs can not only alleviate the problems caused by scale variation, but also decrease the inference time. SSD [14] performs object detection by using the multi-level feature maps from different layers computed in the forward. TDM [15], DSSD [16] and FPN [17] further introduce a top-down pathway and lateral connections to enhance the semantic representation of low-level features at bottom layers. Imperfectly, CNNs are not robust to changes in scale. All aforementioned methods do not take into account the scale mismatch between training and inference. They use the same shared inference module for all scales which decreases detection performance for very small or very large objects.

2.3 Methods for handling mismatch

The mismatch between training and inference will have a great impact on the CNN-based model. Cascade R-CNN [29] proposed a multi-stage object detection framework for the design of high quality object detectors. The cascade R-CNN stages are trained sequentially, using the output of one stage to train the next. The same cascade procedure is applied at inference, enabling a closer match between the hypotheses and the detector quality of each stage. SNIP [18] proposes a scale normalisation method to train objects that fall into the desired scale range during multi-scale training based on the image pyramids strategy. TridentNet [19] generates scale-specific features through multiple parallel branches using dilated convolution to endow their network the same representational power for all objects of different scales. Without applying the image pyramid strategy, 4SD takes a single-scale image as input. Different from TridentNet using dilated convolution to generate scale-specific feature maps, 4SD predict objects of a specific scale range separately by using feature maps of different sizes. Furthermore, SNIP and TridentNet are two-stage detectors while 4SD is a single-stage detector. The detection speed of 4SD is significantly faster than that of SNIP and TridentNet.

3 Proposed method

Our goal is to predict objects of a specific scale range separately by using feature maps of different sizes. We propose an scale specific single shot architecture to achieve this as shown in Fig. 1. We choose three branches as our default setting, detailed in Section 4.3. Our method takes a single-scale image as input. The base network is employed to obtain the cascade input (F_2 , F_1 , and F_0) for SSIM. An SSIM is constructed to generate a parallel multi-branch architecture with feature maps of different sizes (P_S , P_M , and P_L). The desired scale object is predicted by the branch of feature maps of specific sizes, followed by a non-maximum suppression step to produce the final results. To solve the scale mismatch between training and inference, an SSTS is proposed to specialise each branch by sampling object instances of proper scales for training. The construction of our model involves a base network, an SSIM, and a non-maximum suppression module, as introduced in the following.

3.1 Base network

The 4SD is based on Darknet-53 [27]. Darknet-53 uses successive 3×3 and 1×1 convolutional layers and has multiple shortcut connections (residual blocks) as well. It has 53 convolutional layers totally. This network is much more powerful than VGG-16 [30] but still more efficient than ResNet-101 [31]. In order to meet our needs, the base network is constructed by removing the classification layers pre-trained on ImageNet [7]. We set n to represent the reduction rate of the feature maps relative to the input image. Then we use F_2 , F_1 , and F_0 to represent the topmost feature maps that satisfy $n = 8, 16$, and 32 , respectively.

3.2 Scale specific inference module

The SSIM consists of three parallel branches in which each extends from the same topmost feature maps F_0 . The detail of the module is shown in Fig. 2. In detection branch of large object ($SSIM_{(L)}$), a 3×3 convolution is applied prior to a 1×1 convolution to change the number of channels. The 1×1 convolution produces the feature maps P_L with $3 \times (4 + 1 + N)$ channels for the three default boxes at each scale, four bounding box offsets, one objectness prediction, and N object categories, which generates the scores for object classes and shape offsets relative to the default box coordinates. To construct the detection branch of the middle object ($SSIM_{(M)}$), the higher resolution features are required by upsampling spatially coarser. In order to reduce the scale of parameters, we use a 1×1 convolution to reduce the number of channels. To match the dimensions we need, the linear interpolation is used to enlarge the feature maps F_0 . These feature

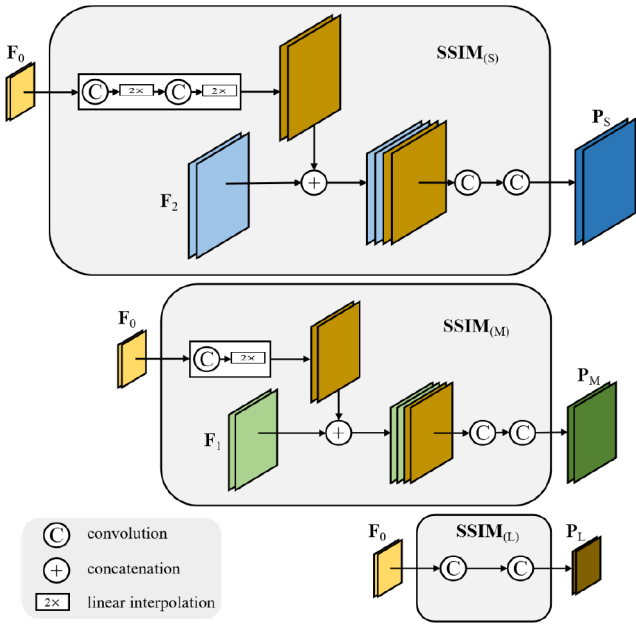


Fig. 2 SSIM for each branch (detailed Section 3.2)

maps are then enhanced with feature maps F_1 via concatenation. Then, we add a series convolution layer after the concatenation to ensure the discriminability of feature maps P_M for detection. As for detection branch of small object (SSIM_(S)), the feature maps F_0 will upsample twice to match the dimensions with F_2 . After a series convolution layer, feature maps P_S were produced for small object detection.

3.3 Scale specific training scheme

The proposed SSIM generates scale specific feature maps based on different sizes. However, CNNs are not robust to changes in scale. For every single branch, the scale mismatch between training and inference still exists. Therefore, it is natural to train objects of different scales on different branches. Here, we propose an SSTS to increase the scale awareness of each branch and avoid training in mismatched branches.

Similar to SNIP [18] and TridentNet [19], we define a valid range $[e_i, k_i]$ for each branch i . During training, we only select the ground truth boxes whose area fall in the corresponding valid range of each branch. Formulaically, for a ground truth bounding box with width w and height h , it is valid for the branch i when

$$e_i^2 \leq wh \leq k_i^2. \quad (1)$$

When adopting an SSTS, ground truth boxes are matched to the branch by falling in the corresponding valid range of each branch. Different from TridentNet, we run k-means clustering on the dimensions of ground truth bounding boxes to get good default boxes for each branch. In our experiments, we predict three boxes at each branch. For example, in COCO dataset, in detection branch of small object, the three clusters were (10×13) , (16×30) , (33×23) . In branch of middle and large objects, that were (30×61) , (62×45) , (59×119) and (116×90) , (156×198) , (373×326) , respectively. To make the model more robust to various input object sizes and shapes, we use multi-scale training, lots of data augmentation, batch normalisation, all the standard stuff.

According to YOLO [10], for each branch i , we optimise the following, multi-part loss function L_i :

$$L_i = L_{loc}^i + L_{con}^i + L_{cls}^i, \quad (2)$$

where L_{loc}^i was a L_1 loss function to regress a candidate bounding box into a target bounding box, L_{con}^i and L_{cls}^i were classic cross-entropy loss for the confidence of box contains an object and class

predictions appearing in the box, respectively. During training, the loss function of all N branches was added up to total loss L_{total} :

$$L_{total} = \sum_{i=1}^N L_i. \quad (3)$$

Finally, those samples which fall outside the valid range, their gradients are set to zero.

3.4 Non-maximum suppression module

During inference, we generate detection results for all branches and then filter out the boxes which fall outside the valid range of each branch. We then use soft-NMS [32] to combine the detection outputs of three branches to obtain the final results.

4 Experiments

4.1 Datasets

We evaluate our method on two common standard datasets. One is the PASCAL VOC [21] and the other is the MS COCO [22] datasets. Both VOC 2007 and VOC 2012 datasets contain 20 object categories. VOC 2007 has a total of 9963 labelled images, of which 5011 are divided into *trainval* sets, and the remaining are *test* sets. We unite the sets of VOC 2012 to train together and finally evaluate on the *test* set of VOC 2007. MS COCO includes 80 object categories. We train using the union of 80k train images and a 35k subset of Val images (*trainval35k* [33]), and evaluate on a set of 20k test images (*test-dev*).

4.2 Implementation details

Experiments are all based on DarkNet-53 [27]. Following other standard detectors [8, 24], the network backbones are pre-trained on the ImageNet [7] and then fine-tuned on the detection dataset. The input images are resized to 544×544 . Our network is trained in an end-to-end manner. During training, for each image, we retain 150 samples for each branch. For the evaluation of VOC, we use the methodology and tools of Hoiem *et al.* [34]. For MS COCO, we report the standard COCO evaluation metric of average precision (AP) [22] as well as AP_{50}/AP_{75} . We also report COCO-style AP_S , AP_M and AP_L on objects of small, medium and large-scale. The object scales are determined by measuring the number of pixels in the object's segmentation mask, S , as follows: small objects (AP_S): $S < 32^2$; medium objects (AP_M): $32^2 < S < 96^2$; large objects (AP_L): $S > 96^2$.

4.3 Number of branches

We study the choice of the number of branches in 4SD.

Depending on the size of the area, the ground truth boxes set is divided into multiple equal parts. Then we run k-means clustering on the dimensions of ground truth bounding boxes of each part to get good default boxes for each branch. Table 1 shows the detection results on the COCO *minival* set [22] using one to four branches.

We observe that using three branches further improves the performance of the detector. Particularly, it brings additional improvements for small objects by 9.9% and for middle objects by 6%. This shows that the effects of 4SD architecture for multi-scale detection. As can be noticed, from two branches to four branches, mAP is slightly dropped step by step for large objects. This result is caused by over-fitting. First, the multi-branch structure introduces several times parameters which may potentially incur over-fitting. Secondly, the SSTS design prevents each branch from training objects of extreme scales, but may also bring about over-fitting problem in each branch caused by reduced effective samples. As shown in Table 1, four branches do not bring further improvement over three branches. Thus, considering the complexity and performance, we choose three branches as our default setting.

Table 1 Detection results on the COCO *minival* set

Branches	Backbone	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
1	DarkNet-53	31.6	50.7	34.2	10.2	34.3	49.6
2	DarkNet-53	34.1	55.7	36.9	16.4	38.6	49.8
3	DarkNet-53	37.2	58.8	40.1	20.1	40.3	49.2
4	DarkNet-53	36.9	58.6	39.9	20.1	40.1	49.0

Bold values indicate the best results of the current experimental group.

Table 2 Detection results on PASCAL VOC dataset

Method	Backbone	mAP	Aero	Bike	Bird	Boat	Bottle	Bus	Car	Cat	Chair
<i>two-stage:</i>											
faster R-CNN [8]	VGG-16	73.2	76.5	79.0	70.9	65.5	52.1	83.1	84.7	86.4	52.0
faster R-CNN [31]	ResNet-101	76.4	79.8	80.7	76.2	68.3	55.9	85.1	85.3	89.8	56.7
ION [33]	VGG-16	75.6	79.2	83.1	77.6	65.6	54.9	85.4	85.1	87.0	54.4
r-FCN [9]	ResNet-101	80.5	79.9	87.2	81.5	72.0	69.8	86.8	88.5	89.8	67.0
<i>single-stage:</i>											
SSD512* [14]	VGG-16	79.5	84.8	85.1	81.5	73.0	57.8	87.8	88.3	87.4	63.5
SSD513 [16]	ResNet-101	80.6	84.3	87.6	82.6	71.6	59.0	88.2	88.1	89.3	64.4
DSSD513 [16]	ResNet-101	81.5	86.6	86.2	82.6	74.9	62.5	89.0	88.7	88.8	65.2
4SD	Darknet-53	83.1	88.7	91.8	83.4	73.2	77.4	90.7	91.8	89.6	68.4

Bold values indicate the best results of the current experimental group.

Method	Cow	Table	Dog	Horse	mbike	Person	Plant	Sheep	Sofa	Train	TV
<i>two-stage:</i>											
faster R-CNN [8]	81.9	65.7	84.8	84.6	77.5	76.7	38.8	73.6	73.9	83.0	72.6
faster R-CNN [31]	87.8	69.4	88.3	88.9	80.9	78.4	41.7	78.6	79.8	85.3	72.0
ION [33]	80.6	73.8	85.3	82.2	82.2	74.4	47.1	75.8	72.7	84.2	80.4
r-FCN [9]	88.1	74.5	89.8	90.6	79.9	81.2	53.7	81.8	81.5	85.9	79.9
<i>single-stage:</i>											
SSD512* [14]	85.4	73.2	86.2	86.7	83.9	82.5	55.6	81.7	79.0	86.6	80.0
SSD513 [16]	85.6	76.2	88.5	88.9	87.5	83.0	53.6	83.9	82.2	87.2	81.3
DSSD513 [16]	87.0	78.7	88.2	89.0	87.5	83.7	51.1	86.3	81.6	85.7	83.7
4SD	86.8	74.2	88.6	90.7	87.8	89.4	57.4	87.2	75.4	87.3	81.4

Bold values indicate the best results of the current experimental group.

Table 3 Results comparison between DSSD513 and 4SD on PASCAL VOC dataset

	mAP	Aero	Bike	Bird	Boat	Bottle	Bus	Car	Cat	Chair	Cow	Table	Dog	Horse	Mbike	Person	Plant	Sheep	Sofa	Train	TV
DSSD513 [16]	81.5	86.6	86.2	82.6	74.9	62.5	89.0	88.7	88.8	65.2	87.0	78.7	88.2	89.0	87.5	83.7	51.1	86.3	81.6	85.7	83.7
4SD	83.1	88.7	91.8	83.4	73.2	77.4	90.7	91.8	89.6	68.4	86.8	74.2	88.6	90.7	87.8	89.4	57.4	87.2	75.4	87.3	81.4
absolute difference	1.6	2.1	5.6	0.8	-1.7	14.9	1.7	3.1	0.8	3.2	-0.2	-4.5	0.4	1.7	0.3	5.7	6.3	0.9	-6.2	1.6	-2.3
relative difference	1.96	2.42	6.49	0.97	-2.26	23.8	1.91	3.49	0.90	4.90	-0.22	-5.71	0.45	1.91	0.34	6.81	12.3	1.04	-7.59	1.86	-2.74

Bold values indicate the best results of the current experimental group.

4.4 PASCAL VOC

We trained our model on the union of 2007 *trainval* and 2012 *trainval*, and tested on the VOC 2007 *test* set. We set the valid range of three branches as [0, 64], [64, 128] and [128, ∞], respectively. The input images are resized to 544 × 544. By default, we train in a batch size of eight on two NVIDIA 1080Ti GPUs. We train 30 epochs in total, with learning rate starting from 10⁻⁴. For the first 20 epochs, we only train the SSIM by freezing all the weights of the original base network. After training for 20 epochs, the learning rate was reduced to 10⁻⁶, and all network parameters were updated.

Table 2 shows our results on the PASCAL VOC2007 test detection. SSD512* is the latest SSD results with data augmentation trick, which are already better than many other detectors. By replacing the base network to Residual-101, which has 101 convolutional layers, the performance has about 1% better than VGG-16. DSSD513 is consistently about 1% better than

SSD513 by adding the deconvolution layers and skip connections. Applying SSTS, 4SD achieves 83.1% mAP better than DSSD513 by 1.6% with shallower layers. Comparing to the two-stage methods, 4SD performs better than most of them. Notably 4SD is much better than other methods which try to consider context information such as ION [33].

We compare 4SD with DSSD513 in Table 3. DSSD513 is based on Residual-101, which has 101 convolutional layers, is much deeper than 4SD. To introduce additional large-scale context in object detection and improve accuracy, especially for small objects, DSSD513 adds the extra deconvolution layers and skip connections structure. In Table 3, the absolute difference is calculated by AP of 4SD minus DSSD513. Relative difference is the ratio of absolute difference to AP of DSSD513. Notably, 4SD shows a large improvement for small objects in test tasks, such as bottle, plant. This shows that better performance is achieved for small object detection with SSIM. However, it has comparatively

Table 4 Detection speed on PASCAL VOC dataset

Method	Backbone	Input resolution	Boxes	FPS	mAP
<i>two-stage:</i>					
faster R-CNN [8]	VGG-16	$\sim 1000 \times 600$	300	5	73.2
faster R-CNN [31]	ResNet-101	$\sim 1000 \times 600$	300	2.4	76.4
ION [33]	VGG-16	$\sim 1000 \times 600$	4000	1.3	79.2
r-FCN [9]	ResNet-101	$\sim 1000 \times 600$	300	9	80.5
<i>single-stage:</i>					
YOLOv2 [26]	Darknet-19	544×544	845	40	78.6
SSD512* [14]	VGG-16	512×512	24,564	19	79.5
DSSD513 [16]	ResNet-101	513×513	43,688	6	81.5
4SD	Darknet-53	544×544	18,207	28	83.1

Table 5 Detection results on the COCO *test-dev* set

Method	Backbone	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
<i>two-stage:</i>							
faster R-CNN [8]	VGG-16	21.9	42.7	—	—	—	—
faster R-CNN [31]	ResNet-101	34.9	55.7	37.4	15.6	38.7	50.9
ION [33]	VGG-16	23.6	43.2	23.6	6.4	24.1	38.3
FPN [17]	ResNet-101	36.2	59.1	39.0	18.2	39.0	48.2
TDM [15]	ResNet-101	36.8	57.7	39.2	16.2	39.8	52.1
SNIPER [18]	ResNet-101	46.1	67.0	51.5	29.6	48.9	58.1
TridentNet [19]	ResNet-101	48.4	69.7	53.5	31.8	51.3	60.3
<i>single-stage:</i>							
SSD300* [14]	VGG-16	25.1	43.1	25.8	6.6	25.9	41.4
SSD321 [16]	ResNet-101	28.0	45.4	29.3	6.2	28.3	49.3
DSSD321 [16]	ResNet-101	28.0	46.1	29.2	7.4	28.1	47.6
SSD512* [14]	VGG-16	28.8	48.5	30.3	10.9	31.8	43.5
SSD513 [16]	ResNet-101	31.2	50.4	33.3	10.2	34.5	49.8
DSSD513 [16]	ResNet-101	33.2	53.3	35.2	13.0	35.4	51.1
YOLOv2 [26]	Darknet-19	21.6	44.0	19.2	5.0	22.4	35.5
YOLOv3 [27]	Darknet-53	33.0	57.9	34.4	18.3	35.4	41.9
4SD	Darknet-53	36.9	58.0	39.7	20.3	39.5	49.7

Bold values indicate the best results of the current experimental group.

worse performance on large-size objects, such as sofa, table. We judge that this is caused by over-fitting discussed in Section 4.3.

The detection speed of 4SD and other methods is presented in Table 4. We evaluate 4SD with batch size 1 on a machine with NVIDIA 1080Ti GPU. As shown in Table 3, the two-stage methods, which uses larger input size (i.e. $\sim 1000 \times 600$), is slower than the single-stage methods. Benefiting from shallower network layers, SSD512* and YOLOv2 are faster than 4SD, but their accuracy are 3.3 and 4.5% worse than ours. 4SD processes an image in 36 ms (28 FPS), with an input size of 544×544 , while 167 ms (6 FPS) is required on DSSD513. In summary, 4SD achieves the real-time speed and best trade-off between accuracy and speed.

4.5 MS COCO

To evaluate our method on a more challenging dataset, we also evaluate 4SD on MS COCO [22]. The valid range of three branches are set to [0, 32], [32, 96], and [96, ∞], respectively. We train the network with a batch size of 8. The model is trained for 60 epochs in total. Freezing all the weights of the base network, we train the SSIM with learning rate at 10^{-4} for the first 30 epochs, then decreased by a factor of 0.1. In the last 10 epochs, the learning rate was reduced to 10^{-6} , and all the parameters were updated.

Table 5 shows the results on MS COCO *test-dev* set. Comparing to the two-stage methods, we see that 4SD is already better than Faster R-CNN (AP 36.9 versus 34.9) and ION (AP 36.9 versus 23.6). We also observe that 4SD has a better improvement on small objects than several detectors applying skip connections structure, e.g. FPN (AP_S 20.3 versus 18.2) and TDM (AP_S 20.3 versus 16.2). Other two-stage methods, e.g. SNIPER and TridentNet, also handle

the mismatch between training and inference. Based on a complex region refinement process and large input size, they achieve better results than 4SD.

The performance of 4SD is superior to most of the single-stage detectors. The 4SD shows improvement on all sizes of objects and achieves 36.9% mAP, which is 3.9% better than YOLOv3 [27] with the same base network. According to this observation, we believe that the SSIM will significantly improve accuracy to detect objects at vastly different scales. We also observe that 4SD increases over DSSD513 by 7.3 points of AP_S (20.3 versus 13.0) for small objects. But the result of AP_L for large objects is a little worse (49.7 versus 51.1), which is due to over-fitting discussed in Section 4.3.

5 Conclusions

In this paper, we propose a novel single-stage method 4SD using feature maps of different sizes to predict objects that fall into the desired scale range. An SSIM is constructed to generate a parallel multi-branch architecture with feature maps of different sizes. The SSTS is designed to increase the scale awareness of each branch and avoid extreme-scale training in unmatched branches. Experimental results on the PASCAL VOC and COCO dataset demonstrated that 4SD will significantly improve accuracy to detect objects at vastly different scales, especially for small objects, while still preserving comparable speed to other detectors.

6 References

- [1] Adelson, E.H., Anderson, C.H., Bergen, J.R., *et al.*: 'Pyramid methods in image processing', *RCA engineer*, 1984, **29**, (6), pp. 33–41

- [2] Dalal, N., Triggs, B.: 'Histograms of oriented gradients for human detection'. 2005 IEEE Computer Society Conf. on Computer Vision and Pattern Recognition (CVPR, 2005), San Diego, CA, USA, June 20-26, 2005
- [3] Lowe, D.G.: 'Distinctive image features from scale-invariant keypoints', *Int. J. Comput. Vis.*, 2004, **60**, pp. 91-110
- [4] Felzenszwalb, P.F., Girshick, R.B., McAllester, D., *et al.*: 'Object detection with discriminatively trained part-based models', *IEEE Trans. Pattern Anal. Mach. Intell.*, 2010, **32**, pp. 1627-1645
- [5] LeCun, Y., Boser, B., Denker, J.S., *et al.*: 'Backpropagation applied to handwritten zip code recognition', *Neural Comput.*, 1989, **1**, pp. 541-551
- [6] Krizhevsky, A., Sutskever, I., Hinton, G.: 'Imagenet classification with deep convolutional neural networks'. Neural Information Processing Systems (NIPS, 2012), Harrah's and Harveys Lake Tahoe, NV, USA, December 3-8, 2012
- [7] Deng, J., Dong, W., Socher, R., *et al.*: 'Imagenet: a large-scale hierarchical image database'. IEEE Conf. on Computer Vision and Pattern Recognition, 2009. CVPR, 2009, Miami, Florida, USA, June 20-25, 2009, pp. 248-255
- [8] Ren, S., He, K., Girshick, G., *et al.*: 'Faster R-CNN: towards real-time object detection with region proposal networks'. Neural Information Processing Systems (NIPS, 2015), Montreal, Quebec, Canada, December 7-12, 2015
- [9] Dai, J., Li, Y., He, K., *et al.*: 'R-FCN: object detection via region-based fully convolutional networks'. NIPS'16 Proc. of the 30th Int. Conf. on Neural Information Processing Systems, Barcelona, Spain, December 5-10, 2016, pp. 379-387
- [10] Redmon, J., Divvala, S., Girshick, R., *et al.*: 'You only look once: unified, real-time object detection'. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR, 2016), Las Vegas, NV, USA, June 27-30, 2016
- [11] He, K., Zhang, X., Ren, S., *et al.*: 'Spatial pyramid pooling in deep convolutional networks for visual recognition'. European Conf. on Computer Vision (ECCV, 2014), Zurich, Switzerland, September 6-12, 2014
- [12] Girshick, R.: 'Fast R-CNN'. Int. Conf. on Computer Vision (ICCV, 2015), Santiago, Chile, December 11-18, 2015
- [13] Huang, Z., Huang, L., Gong, Y., *et al.*: 'Mask scoring R-CNN'. Computer Vision and Pattern Recognition (CVPR, 2019), Long Beach, CA, USA, June 16-20, 2019
- [14] Liu, W., Anguelov, D., Erhan, D., *et al.*: 'SSD: single shot multibox detector'. European Conf. on Computer Vision (ECCV, 2016), Amsterdam, The Netherlands, October 8-16, 2016
- [15] Shrivastava, A., Sukthankar, R., Malik, J., *et al.*: 'Beyond skip connections: top-down modulation for object detection', arXiv preprint arXiv:1612.06851, 2016
- [16] Fu, C.Y., Liu, W., Ranga, A., *et al.*: 'DSSD: deconvolutional single shot detector', arXiv preprint arXiv:1701.06659, 2017
- [17] Lin, T.Y., Dollar, P., Girshick, R.B., *et al.*: 'Feature pyramid networks for object detection'. 2017 IEEE Conf. on Computer Vision and Pattern Recognition (CVPR, 2017), Honolulu, Hawaii, USA, July 21-26, 2017
- [18] Singh, B., Davis, L.S.: 'An analysis of scale invariance in object detection-SNIP'. Computer Vision and Pattern Recognition (CVPR, 2018), Salt Lake City, Utah, USA, June 18-22, 2018
- [19] Li, Y., Chen, Y., Wang, N., *et al.*: 'Scale-aware trident networks for object detection'. ICCV, 2019, Seoul, Korea, October 27 - November 2, 2019
- [20] Yu, F., Koltun, V.: 'Multi-scale context aggregation by dilated convolutions'. ICLR, 2016, San Juan, Puerto Rico, May 2-4, 2016
- [21] Everingham, M., Gool, L.V., Williams, C.K.I., *et al.*: 'The PASCAL visual object classes (VOC) challenge', *Int. J. Comput. Vis.*, 2010, **88**, (2), pp. 303-338
- [22] Lin, T.Y., Maire, M., Belongie, S., *et al.*: 'Microsoft COCO: common objects in context'. European Conf. on Computer Vision, Zurich, Switzerland, September 6-12, 2014, pp. 740-755
- [23] Sermanet, P., Eigen, D., Zhang, X., *et al.*: 'Overfeat: integrated recognition, localization and detection using convolutional networks'. Proc. Int. Conf. Learning Representations (ICLR), 2014, Banff, Canada, April 14-16, 2014
- [24] Girshick, G., Donahue, J., Darrell, T., *et al.*: 'Rich feature hierarchies for accurate object detection and semantic segmentation'. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR, 2014), Columbus, Ohio, USA, June 23-28, 2014
- [25] Uijlings, J.R., van de Sande, K.E., Gevers, T., *et al.*: 'Selective search for object recognition', *Int. J. Comput. Vis.*, 2013, **104**, pp. 154-171
- [26] Redmon, J., Farhadi, A.: 'YOLO9000: better, faster, stronger'. CVPR, 2017, Honolulu, Hawaii, USA, July 21-26, 2017
- [27] Redmon, J., Farhadi, A.: 'YOLOv3: an incremental improvement', arXiv preprint arXiv:1804.02767, 2018
- [28] Lin, T., Goyal, P., Girshick, R., *et al.*: 'Focal loss for dense object detection'. ICCV, 2017, Venice, Italy, October 22-29, 2017
- [29] Cai, Z., Vasconcelos, N.: 'Cascade R-CNN: delving into high quality object detection'. CVPR, 2018, Salt Lake City, Utah, USA, June 18-22, 2018
- [30] Simonyan, K., Zisserman, A.: 'Very deep convolutional networks for largescale image recognition', ICLR, 2015, San Diego, CA, USA, May 7-9, 2015
- [31] He, K., Zhang, X., Ren, S., *et al.*: 'Deep residual learning for image recognition'. 2016 IEEE Conf. on Computer Vision and Pattern Recognition (CVPR, 2016), Las Vegas, NV, USA, June 27-30, 2016
- [32] Bodla, N., Singh, B., Chellappa, R., *et al.*: 'SoftNMS improving object detection with one line of code'. IEEE Int. Conf. on Computer Vision (ICCV, 2017), Venice, Italy, October 22-29, 2017
- [33] Bell, S., Zitnick, C.L., Bala, K., *et al.*: 'Inside-outside net: detecting objects in context with skip pooling and recurrent neural networks'. CVPR, 2016, Las Vegas, NV, USA, June 27-30, 2016
- [34] Hoiem, D., Chodpathumwan, Y., Dai, Q.: 'Diagnosing error in object detectors'. Computer Vision ECCV, 2012, Florence, Italy, October 7-13, 2012, pp. 340-353